Daniel English-Brown

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Summary

Data-driven MSc Theoretical Physics student at King's College London with advanced expertise in Python, quantitative modelling, and statistical analysis. Experienced building valuation and optimisation algorithms for financial instruments, developing full-stack data applications, and presenting complex results clearly. Seeking quantitative research or data-science roles applying rigorous mathematical reasoning and coding discipline to real-world markets or analytics problems.

Experience

Quantitative Research Intern, Perbak Capital Partners (London) Sept 2024

- Built and validated Black–Scholes and binomial pricing models in Python for derivative valuation and volatility estimation.
- Designed convex-optimisation pipelines in CVXPY to construct market-neutral portfolios, reducing simulated drawdowns by 15 %.
- Implemented statistical diagnostics and volatility clustering detection on non-Gaussian return distributions.
- Automated financial-time-series cleaning and feature-engineering scripts for faster data prep.

Lead Web Developer / Data Engineer, Expendid Sept 2025–Present

- Designed backend Flask APIs and automated analytics dashboards for an AI-driven finance-management tool.
- Implemented dynamic pricing-model testing using historical expenditure data; improved accuracy of projections by 12 %.
- Managed Git-based development workflow and continuous-deployment pipeline across staging and production.

Independent Data Projects (Portfolio) 2023–2025

- Created Python-based risk-analysis scripts exploring portfolio optimisation via mean-variance and Sharpe maximisation.
- Analysed UK market data to back-test option-strategy profitability; visualised results with Matplotlib and Seaborn.
- Built and hosted interactive dashboards on danielenglish.me.

STEM Outreach Ambassador, King's College London Sept 2024–Present

- Delivered physics demonstrations and analytics workshops introducing data modelling concepts to students.

Technical Projects

Finite-Difference Time-Domain Electromagnetic Simulator (Python)

Developed numerical solver implementing Yee-lattice updates to compute resonant modes of dielectric rings. Produced quantitative validation of Q-factors and spectral spacing; results visualised via Matplotlib.

Expendid Finance Platform —Analytics Module

Built forecasting engine and user-metrics dashboard linking Flask backend with SQL-style data store; supports A/B testing of algorithmic improvements.

Portfolio Optimisation Toolkit

End-to-end notebook applying CAPM inputs, covariance matrices and CVXPY to generate efficient frontiers.

Education

2025–2026	**MSc Theoretical Physics**, King's College London —Focus on Computational Methods and Applied Mathematics
2022–2025	**BSc (Hons) Theoretical Physics**, King's College London
2020–2022	**A-Levels**, Ampleforth College —Physics, Chemistry, Mathematics, French (AABB); Academic Scholarship

Technical Skills

Programming	Python (NumPy, Pandas, SciPy, Matplotlib, CVXPY), MATLAB, JavaScript, SQL, Git
Quantitative Modelling	Option Pricing, Portfolio Optimisation, Risk Analytics, Monte Carlo Simulation
Statistics & Data	Time-Series Analysis, Regression, Hypothesis Testing, Feature Engineering, Data Visualisation
Tools	Flask, Excel, LaTeX, Power BI (Basics)
Languages	English (C2), Mandarin (C1), French (B2)

Achievements

SAT 1510 (99th percentile) | BSc (Hons) Theoretical Physics | HSK 5 (Mandarin Proficiency) | Academic Scholarship —Ampleforth College